

Introduction to Microeconomics
Lecture 4—Comparative Advantage

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Comparative Advantage: The Basis of Exchange

Why do people exchange goods and services in the first place? Why not just produce our own food, cars, clothing, shelter, and the like? The answer is that we can all have more of every good and service if we specialize in the activities at which we are **relatively** most efficient.

Arsen and Batyrbek

Arsen is a house painter whose room needs fixing, while Batyrbek is a roofer whose house needs painting. Although Arsen is a house painter, he also knows how to repair roofs. Batyrbek, for his part, knows how to paint houses.

	Painting	Roofing
Arsen	300 hours	400 hours
Batyrbek	200 hours	100 hours

- Note that Batyrbek has an **absolute advantage** → in each activity, Batyrbek takes less hours than Arsen to perform each task.
- Should Batyrbek do both the roofing and painting jobs for both houses?

Arsen and Batyrbek

	Painting	Roofing
Arsen	300 hours	400 hours
Batyrbek	200 hours	100 hours

- Batyrbek may say that him doing both jobs is unfair. A more telling objection is that it is **inefficient**.
- The reason is that Arsen has a **comparative advantage** over Batyrbek at painting, which means that he is **relatively** more efficient at painting than Batyrbek.
- Key word: **relatively**.
- It means that Arsen has a **lower opportunity cost** at performing the task of painting than Batyrbek has.
- Arsen: the opportunity cost of painting a house is 0.75 roofing jobs. $= \frac{300}{400} = \frac{3}{4}$
- Batyrbek: the opportunity cost of painting a house is 2 roofing jobs. $= \frac{200}{100} = 2$
- Arsen has a comparative advantage at painting, and it makes sense for him to do both painting jobs, leaving the two roofing jobs to Batyrbek.

The principle of Comparative Advantage

Everyone does best when each person (or country) concentrates on the activities in which he or she is relatively most efficient.

Caution!

Pay close attention to the form in which the productivity information is provided. Your goal in each case is to find each person's opportunity cost of producing the good in question.

Aiym and Balzhan

Aiym can milk a goat in 10 minutes and shear a sheep in 4 minutes. Balzhan can milk a goat in 6 minutes and shear a sheep in 3 minutes. Which statement below is true?

- ① Aiym has no comparative advantage.
- ② Balzhan should do both tasks because he has an absolute advantage in both.
- ③ Aiym has a comparative advantage in shearing sheep and Balzhan has a comparative advantage in milking goats.
- ④ Aiym has a comparative advantage in milking goats and Balzhan has a comparative advantage in shearing sheep.
- ⑤ None of the above statements is true.

	C.O. of milking goats	C.O. of shearing
Aiym	$10/4 = 2.5$	$4/10 = 2/5$
Balzhan	$6/3 = 2$	$3/6 = 1/2$

Aiym and Balzhan

Aiym can milk 10 goats per hour or shear 4 sheep per hour. Balzhan can milk 6 goats per hour or shear 3 sheep per hour. Which statement below is true?

- ① Aiym has no comparative advantage.
- ② Balzhan should do both tasks because he has an absolute advantage in both.
- ③ Aiym has a comparative advantage in shearing sheep and Balzhan has a comparative advantage in milking goats.
- ④ Aiym has a comparative advantage in milking goats and Balzhan has a comparative advantage in shearing sheep.
- ⑤ None of the above statements is true.

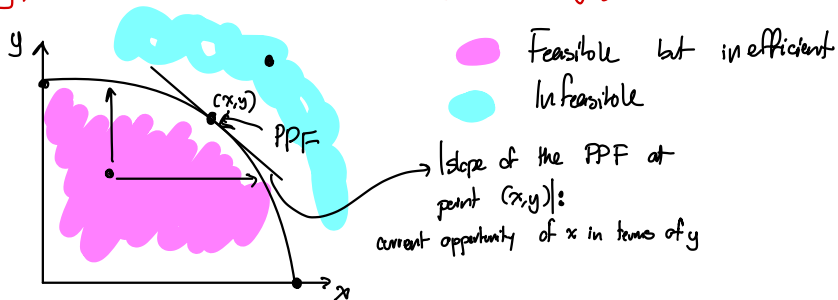
	C.O. of milking	C.O. of shearing sheep
Aiym	$4/10 = 0.4$ sheep sheared	2.5 goats milked
Balzhan	$3/6 = 0.5$ sheep sheared	2 goats milked.

The production possibilities frontier (PPF)

Production Possibilities Frontier

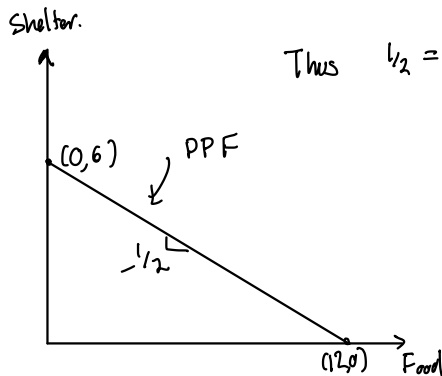
It is the boundary between those combinations of goods and services that can be produced, given current resources, and those that cannot. Any point inside the PPF is **feasible** but **inefficient**. Any point outside the PPF is **infeasible**. Points on the frontier are **efficient**. The slope of the PPF determines the **opportunity cost** of one activity/good vs the other one.

Two activities (x, y) and some limited resources to engage in those act.



Daniyar and the island

Daniyar has been cast away to a remote island, where he is the only inhabitant. Daniyar can produce 6 m²/week of shelter or 12 kg/week of food. Describe Daniyar's PPF.



Thus $\frac{1}{2} =$ cost of opportunity of units of food
in terms of units of shelter foregone.

That is: each extra unit (1 kg) of food
0.5 units (m²) of shelter.

$S = \text{shelter}$

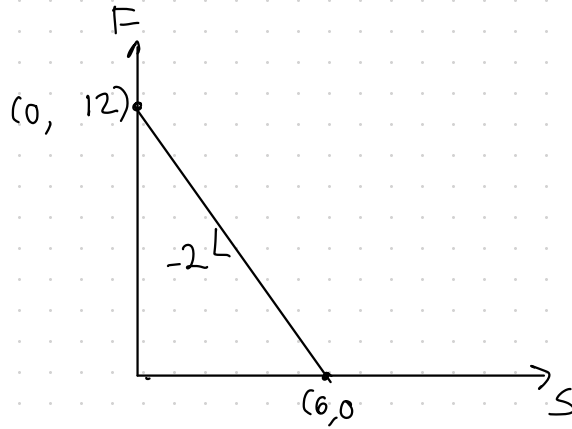
$F = \text{Food}$

$$S - 6 = -\frac{1}{2} (F - 0)$$

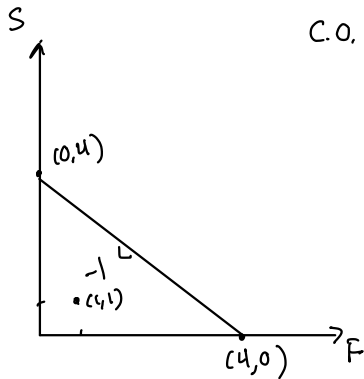
$$S = 6 - \frac{1}{2} F$$

$$S = 6 - \frac{1}{2}F \Rightarrow 2S = 12 - F$$

$$F = 12 - 2S$$



Karim can produce 4 m²/week of shelter ^{OR} ~~4 m²~~ 4 kg/week of food. Describe ~~Karim's~~ ^{Karim} PPF.



C.O. 1 Food costs 1 Shelter

$$S = 4 - F$$

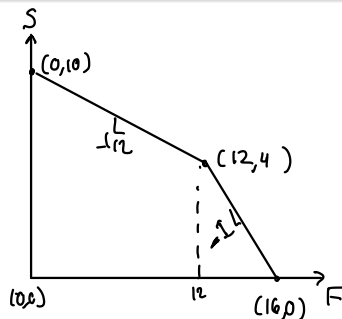
$$\hookrightarrow F = 4 - S$$

PPF in a two-person Economy

In the examples above, Daniyar has a comparative advantage in producing food, while Karim has a comparative advantage producing shelter.

Daniyar and Karim

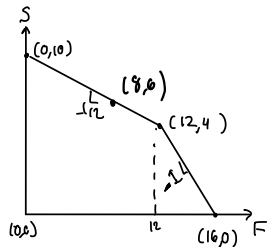
Daniyar can produce 6 m²/week of shelter and 12kg/week of food. Karim can produce 4 m²/week of shelter and 4 m²/week of food. Describe the economy's PPF.



$$S = \begin{cases} -\frac{1}{2}(F-0) + 10 & 0 \leq F \leq 12 \\ \quad \quad \quad \hookrightarrow 10 - F/2 \\ -1(F-12) + 4 & 12 < F \leq 16 \\ \quad \quad \quad \hookrightarrow 16 - F \end{cases}$$

Daniyar and Karim

Suppose Daniyar and Karim decide to join forces and have decided to consume, jointly, 6 m²/week of shelter and 8 kg/week of food. How should they divide the task of producing these quantities?



$$S = \begin{cases} -\frac{1}{2}(F-0) + 10 & 0 \leq F \leq 12 \\ \quad \quad \quad \hookrightarrow 10 - F/2 \\ -1(F-0) + 16 & 12 < F \leq 16 \\ \quad \quad \quad \hookrightarrow 16 - F \end{cases}$$

producing

Karim is (0, 4)

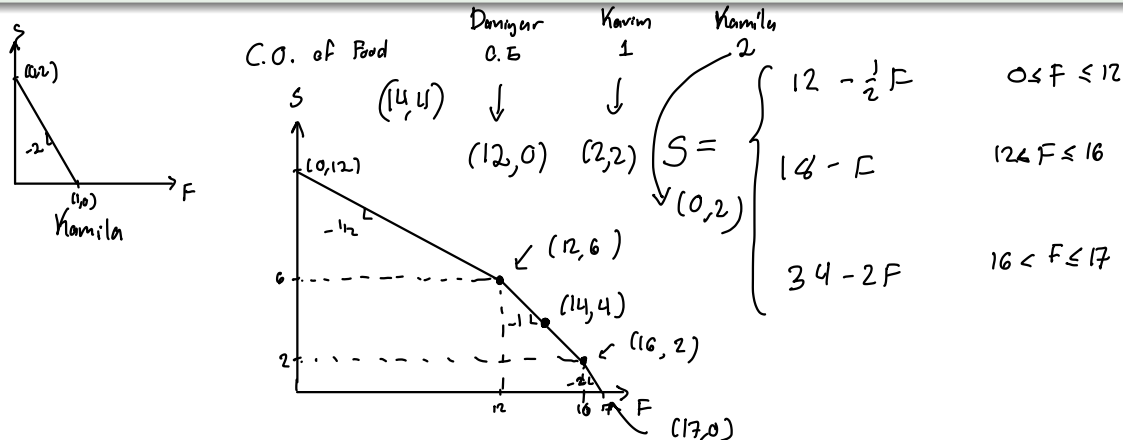
Daniyar is (8, 2)

The Principle of Increasing Opportunity Cost

In expanding the production of any good, first employ those resources with the lowest opportunity cost, and only afterward turn to resources with higher opportunity costs.

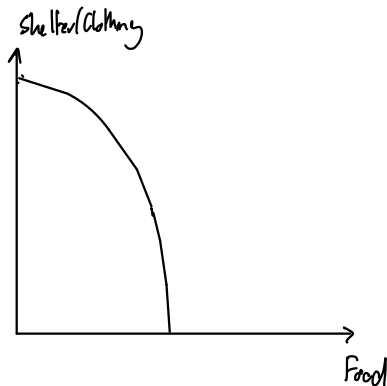
Kamila joins the group of castaways

Suppose that Kamila also becomes stranded in the island. Kamila can produce 2 m²/week of shelter or 1 kg/week of food. Start by describing her individual PPF. Then describe the PPF of the three-person economy. If they jointly decide to produce 14 kg/week of food and 4 m²/week of shelter, how should each person's work time be allocated?



The production possibilities frontier with many workers

Intuition behind its shape: produce the initial units of clothing using the resources that are relatively most efficient at clothing production, and only then turn to those that are relatively less efficient at clothing production.



Add more
and
more
workers

